

Final Project Proposal-Group 04

Project Topic: Predicting Garments Worker Productivity

Dataset: <https://www.kaggle.com/datasets/ishadss/productivity-prediction-of-garment-employees>

Background: The garments manufacturing industry plays a critical role in operational efficiency and economic growth. Worker productivity directly impacts production planning, cost control, and timely delivery. However, productivity is influenced by multiple factors such as work conditions, incentives, workload, and time-related variables, making manual assessment difficult. The Machine learning model and Regression model can help uncover hidden patterns in these factors and support data-driven decision-making.

Description of the Dataset: This project uses the **Garments Worker Productivity Dataset**, which contains daily production-related information from garment factories. This dataset includes 1198 observations with numerical and categorical variables such as team information, targeted productivity, department, overtime, incentives, idle time, and other operational attributes. The target variable represents **actual productivity**, measured as a continuous value.

Objectives:

- Perform exploratory data analysis (EDA) to understand key drivers of productivity.
- Preprocess data using appropriate encoding and scaling techniques.
- Build and compare multiple regression models and Evaluate model performance using suitable metrics.
- Identify the most influential factors affecting productivity.
- Predict the productivity of the garment works using the fitted model.

Methodology:

- **Data Preprocessing:** Handle missing values, encode categorical variables, and scale numerical features using pipelines.
- **Exploratory Analysis:** Analyze correlations and feature distributions, cluster analysis (cluster wise explanatory analysis).
- **Model Development:** Train the both Regression and Machine learning model on the tarin data.
- **Model Evaluation:** Use R^2 and RMSE on training and test datasets, along with cross-validation.
- **Model Interpretation:** Analyze feature importance and coefficients.

Conclusion: This project aims to demonstrate how Regression and machine learning model can be effectively applied to real-world industrial data to improve productivity predicting and decision-making in the garments manufacturing sector.

Prepared by:

P.G. Dinithi Oshadhi -s16713

P.V.K. Fernando-s16919

A.M. Chamika Dilshan: s16927